Q1) Range Sum Queries

You are given an array of N integers. You will be asked Q queries. Each query provides two indices L and R. For every query, compute the sum of all elements in the array from index L to R.

Q2) Longest Subarray with Sum $\leq X$

Given an array of positive integers and a target value X, find the length of the longest contiguous subarray whose sum does not exceed X.

Q3) Factory Machines

You have N machines, where the i-th machine takes t_i time to produce one product. All machines work simultaneously. Find the minimum time required to produce exactly T products.

Q4) Sum of Pairwise XOR

Given an array of N integers, compute the sum of XOR over all pairs:

$$\sum_{1 \leq i < j \leq N} (A[i] \oplus A[j])$$

Q5) Interval Scheduling

You are given N intervals (s_i, e_i). You can attend only one event at a time. Find the maximum number of non-overlapping intervals you can attend.

Advanced Topic 1: Ternary Search

Given a continuous unimodal function $f(x)$ defined over $[L, R]$, find the value of x that maximizes (or minimizes) $f(x)$.

Advanced Topic 2: Maximum Subarray Median

You are given an array of N integers and an integer K. Find the maximum possible median of any contiguous subarray of length at least K.